Intelligent Electricity Monitoring, Theft Detection, and Consumption Forecasting: Methods, Challenges, and the VoltIQ Hardware-Integrated Framework

Devanshi Mann, Dhritiman Sen, Divyanshu Sharma, Charchit Sharma
Department of Computer Science & Engineering (Data Science)
Swami Keshvanand Institute of Technology, Management & Gramothan
Jaipur, Rajasthan, India
Project Mentor: Harpreet Singh Gill

***Abstract*—Electricity theft and unmonitored consumer inefficiencies inflict severe financial deficits on electric utility corporations and disrupt generation scheduling. Traditional electromechanical and static meters lack automated tamper-detection capabilities and predictive insight into consumer load profiles. This review paper provides a rigorous systematic survey of contemporary literature spanning three interconnected domains: Internet-of-Things (IoT) sensing hardware, machine learning (ML) and deep learning (DL) algorithms for non-technical loss (NTL) identification, and hybrid time-series load forecasting. A comparative analysis of thirteen peer-reviewed studies published across IEEE Xplore, ScienceDirect, IET, and Springer between 2018 and 2025 is conducted. We analyze supervised ensemble classifiers (XGBoost, LightGBM, CatBoost), class-rebalancing mechanisms (SMOTE), deep spatial-temporal feature extractors (CNN-LSTM, Wide & Deep CNN), unsupervised anomaly detectors (Isolation Forest, Canopy-KMeans), and hybrid forecasters (Prophet-BO-XGBoost). The literature reveals critical research gaps: existing studies predominantly evaluate software algorithms on static offline benchmarks (such as the State Grid Corporation of China dataset) without physical edge validation, whereas existing hardware prototypes remain confined to passive telemetry without automated load isolation, explainability, or integrated predictive modeling. To address these limitations, this paper synthesizes a unified, closed-loop architectural framework termed VoltIQ, which harmonizes dual-sensor edge acquisition (ESP32, ZMPT101B, SCT-013), offline-resilient local buffering, unsupervised anomaly screening, SMOTE-boosted gradient boosting, Prophet-based demand baseline forecasting, and a multi-condition persistent confirmation rule triggering a high-power cutoff relay.**

***Index Terms*—Electricity theft detection, smart metering, ESP32, non-technical losses (NTL), XGBoost, SMOTE, Isolation Forest, Prophet forecasting, load profiling, IoT relay control, SDG 7.**

I. Introduction

Non-Technical Losses (NTLs)—predominantly driven by illicit electricity tapping, meter tampering, physical bypasses, and cyber-spoofing in advanced metering infrastructure (AMI)—account for tens of billions of dollars in lost global utility revenues annually [1], [2]. Beyond economic depletion, unauthorized load drawing causes localized distribution transformer overloads, voltage sags, power grid instability, and heightened fire hazards [3]. Historically, distribution system operators (DSOs) relied on periodic manual inspections and physical auditing, an operational model that is labor-intensive, logistically slow, and easily circumvented by intermittent fraud [4]. The advent of the smart grid paradigm has motivated automated, data-driven approaches; however, utilities continue to grapple with multifaceted technological hurdles [1], [5].

From a computational and systems standpoint, three fundamental difficulties impede the deployment of reliable theft-detection systems:

1) **Extreme Class Imbalance:** In operational distribution grids, fraudulent consumers constitute a small minority (typically 1% to 8% of the total consumer base). Off-the-shelf classifiers trained on raw smart meter telemetry exhibit high aggregate classification accuracy while suffering near-zero recall on actual theft instances [1], [4]. Addressing this requires tailored oversampling strategies, synthetic attack formulations, or cost-sensitive learning paradigms [1], [6].
2) **Measurement Noise and Hardware Constraints:** Low-cost edge current and voltage sensors (such as split-core current transformers and micro-voltage transformers) are prone to thermal drift, analog-to-digital converter (ADC) non-linearities, and phase displacement, introducing double-digit baseline measurement errors if uncalibrated [12], [13]. These physical anomalies propagate into downstream ML feature pipelines, causing false positives.
3) **Decoupling of Algorithms and Physical Actuation:** The vast majority of published theft-detection research focuses solely on algorithmic optimization using historical benchmark repositories—most notably the State Grid Corporation of China (SGCC) dataset [2], [4], [5]. Conversely, published IoT hardware meters stop at passive telemetry streaming or basic threshold alerts [12], [13]. There is a pronounced absence of closed-loop frameworks that integrate real-time sensing, dynamic baseline forecasting, two-stage anomaly verification, and physical load-isolation actuation [7], [11].

Addressing these interconnected challenges is directly

aligned with United Nations Sustainable Development Goal 7 (Affordable and Clean Energy) by minimizing distribution waste, improving grid resilience, and empowering end-users with real-time consumption transparency.

This paper presents an extensive literature survey across 13 reputable, peer-reviewed publications collected from IEEE Xplore, ScienceDirect (Elsevier), Springer, and IET. We dissect the technical evolution of smart metering hardware, supervised fraud detection, unsupervised anomaly screening, and consumption forecasting. Furthermore, we construct an explicit comparative matrix, formulate identified research gaps, and propose the architectural blueprint of **VoltIQ**—a reference-guided, hardware-integrated smart metering and automated theft mitigation system.

II. Literature Review

The technical landscape of intelligent electricity monitoring and fraud mitigation can be categorized into six complementary dimensions: low-cost IoT edge sensing, supervised ensemble theft classification, class imbalance rebalancing, unsupervised anomaly detection, hybrid consumption forecasting, and deep spatial-temporal/cyber-resilient architectures.

A. Low-Cost IoT Sensing Hardware and Metering Architectures

Edge telemetry acquisition forms the foundation of any smart metering infrastructure. Low-cost microcontrollers with integrated wireless stacks, such as the Espressif ESP32, have democratized high-frequency power data acquisition.

Shinde et al. [12] engineered an IoT-based tamper-proof smart energy meter incorporating current transformers and voltage sensing with an ESP32 microcontroller, demonstrating automated load shedding and tamper notifications dispatched across cloud channels. Their setup validated that edge microcontrollers could execute differential current analysis—monitoring the difference between incoming feeder current and outgoing consumer load current—to instantly uncover physical line tapping. However, their detection scheme was strictly deterministic (static threshold comparison), rendering it unable to catch non-linear consumption tapering or intermittent fraud.

Jayasree et al. [13] evaluated a low-cost IoT energy meter utilizing an ESP32 paired with ZMPT101B micro-voltage transformer modules and ACS712 current sensors. The architecture transmitted voltage, current, and computed power metrics to cloud endpoints via HTTP/MQTT protocols. While demonstrating real-time monitoring affordability, the study documented inherent analog sensing inaccuracies resulting from microcontroller ADC non-linearities and temperature variances. Crucially, the prototype lacked predictive capabilities, multi-variate machine learning analysis, and local persistence buffering during network disconnection.

B. Supervised Ensemble Theft Classification

Supervised learning algorithms model the boundary between legitimate and fraudulent consumption by extracting statistical features from high-frequency or daily load profiles.

Punmiya and Choe [1] introduced the Gradient Boosting Theft Detector (GBTD) in the *IEEE Transactions on Smart Grid*. They evaluated Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Categorical Boosting (CatBoost) on smart meter profiles. Rather than relying purely on raw time series, they applied comprehensive feature engineering—extracting rolling mean, standard deviation, variance, minimum, maximum, and weighted feature importance (WFI). Their empirical findings revealed that XGBoost consistently delivered the highest Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and superior detection rates with minimal false positive rates, establishing tree ensembles as the gold standard for tabular consumption analytics.

Kawoosa et al. [3] proposed an ensemble learning approach in *IET Generation, Transmission & Distribution*, addressing smart meter energy theft by synthesizing six realistic mathematical attack models (partial reduction, mean-shifting, intermittent scaling, random bypass). Evaluating XGBoost, Random Forest, AdaBoost, and Support Vector Machines (SVM), the authors reported a 96% detection rate and a 3% false positive rate using XGBoost. Their work proved that gradient boosting handles non-linear multi-modal usage behavior robustly across diverse consumer clusters.

Pereira and Saraiva [7] investigated machine learning models for Non-Technical Loss detection in *Electric Power Systems Research* (ScienceDirect), conducting empirical evaluations of Random Forest, Extra Trees, and Decision Trees across diverse consumer customer segments. They proved that customer-specific models significantly outperform aggregate global models, because industrial, commercial, and residential sectors display disparate diurnal consumption signatures. However, they observed that residential profiles exhibit the highest error rates due to stochastic human occupancy patterns.

C. Class Imbalance Handling and Resampling Strategies

Because power theft is an anomalous occurrence in real-world distribution networks, classifiers suffer from severe majority-class bias.

Aslam et al. [4] investigated this challenge in *IEEE Access*, demonstrating that standard models trained on the uncurated SGCC dataset achieve misleadingly high raw accuracies ($> 94\%$) while missing more than 70% of fraudulent users. To resolve this, the authors integrated the Synthetic Minority Over-sampling Technique (SMOTE) strictly across the training partition, synthesizing interpolative minority samples in feature space. When paired with Gated Recurrent Units (GRU) and ensemble classifiers, SMOTE improved the minority class recall from 42% to over 89%.

Punmiya and Choe [1] and Kawoosa et al. [3] similarly demonstrated that synthetic minority balancing (either through algorithmic interpolation such as SMOTE or via physically grounded mathematical attack functions) is mandatory to prevent decision boundary distortion toward the majority class.

D. Unsupervised Anomaly Detection in Smart Meter Data

In operational deployments, historical ground-truth theft labels are often missing, unverified, or legally contested, severely limiting the applicability of purely supervised learning.

Razavi et al. [6] addressed this in *Applied Energy* (Elsevier) by constructing a feature-engineering framework for Advanced Metering Infrastructure. They extracted customer-centric consumption features (cyclical daily trends, peak-to-average ratios, seasonal variations) and evaluated both unsupervised clustering and supervised validation. Their framework demonstrated that behavioral feature extraction transforms noisy high-frequency kWh series into structured representations where anomalous behavioral shifts become statistically separable.

Wang et al. [9] developed an unsupervised electricity anomaly detection model in *IEEE Access*, coupling an improved Canopy-KMeans clustering algorithm with the Isolation Forest (iForest) algorithm. Canopy-KMeans was utilized as a high-speed pre-clustering filter to partition consumers into homogeneous behavior clusters, after which Isolation Forest constructed ensemble isolation trees to score consumption residuals. By evaluating how quickly a data point is isolated in random hyperplane partitions, the model detected anomalous deviations without requiring historical fraud labels, achieving superior computational efficiency and a low false alarm rate on large-scale AMI telemetry.

ELhadad et al. [10] proposed a multi-stage anomaly prediction pipeline combining unsupervised Isolation Forest with supervised Decision Trees and Random Forests in the *International Journal of Technology*. The authors utilized Isolation Forest to automatically screen and label unlabelled smart meter load sequences into normal and anomalous partitions, and then trained predictive classifiers on sequential lag features to anticipate anomalies in subsequent time windows. Their work established that unsupervised scoring can effectively bootstrap supervised learning pipelines when labeled datasets are unavailable.

E. Hybrid Time-Series Consumption Forecasting

Accurate baseline load forecasting provides utilities with generation scheduling visibility and establishes a dynamic reference against which anomalous consumption drops can be detected.

Zeng et al. [11] introduced a Prophet-BO-XGBoost forecasting framework in *Energies*. The authors recognized that power load curves exhibit strong calendar periodicity (daily, weekly, seasonal cycles) alongside intricate non-linear dependencies driven by external meteorological and socio-economic variables. Their architecture utilized Facebook Prophet to decompose long-term trend and seasonal components, employed XGBoost to fit the non-linear residuals against exogenous features, and applied Bayesian Optimization (BO) to dynamically tune XGBoost hyperparameters. The hybrid model outperformed standalone LSTM, Support Vector Regression (SVR), and standard ARIMA models, demonstrating that combining additive time-series decomposition with gradient-boosted residual regression provides state-of-the-art accuracy in short-term load forecasting.

F. Deep Learning and Adversarial / Cyber-Resilient Architectures

Advanced attackers frequently utilize physical bypasses or execute cyber-intrusions into smart communication links to alter telemetry streams without triggering abrupt threshold trips.

Zheng et al. [2] developed a seminal Wide & Deep Convolutional Neural Network (CNN) architecture in *IEEE Transactions on Industrial Informatics* validated on the SGCC dataset. The "Deep" 2D-CNN branch learned complex temporal cross-day and intra-day consumption correlations directly from raw periodic matrices, while the "Wide" fully connected component analyzed static socio-demographic features. Their model demonstrated that deep feature representations capture subtle, gradual downward load tapering that escapes standard statistical variance checks.

Hasan et al. [8] proposed a hybrid CNN-LSTM architecture in *Energies* to capture spatial and temporal correlations simultaneously. The CNN layers extracted local invariant features across daily usage windows, while the Long Short-Term Memory (LSTM) recurrent network mapped sequential dependencies over weeks and months. The hybrid deep model outperformed standard multi-layer perceptrons and basic SVMs on imbalanced grid data.

Takiddin et al. [5] addressed cybersecurity vulnerabilities in smart meter data in the *IEEE Transactions on Smart Grid*, analyzing the vulnerability of ML theft detectors against data poisoning and label-flipping attacks. To counteract adversarial data manipulation, they designed a robust ensemble detector integrating Deep Autoencoders with Attention (AEA) and Gated Recurrent Units. Their system verified that autoencoder reconstruction errors act as a reliable barrier against poisoned telemetry, ensuring that stealthy manipulation does not degrade the core theft detection classifier.

III. Comparative Analysis

To provide a comprehensive synthesis of the surveyed literature, Table I details the comparative matrix encompassing all thirteen reviewed studies. Each paper is systematically evaluated across Paper Title, Authors, Publication Year, Methodology / Technique, Key Findings, Advantages, and Limitations.

IV. Research Gaps

A systematic synthesis of the literature reveals four critical research gaps that currently divide theoretical machine learning research from practical electrical engineering deployment:

A. Disconnect Between Algorithmic Benchmarks and Hardware Deployment

The literature is sharply bifurcated. On one hand, advanced machine learning studies [1]–[5], [7], [8] report impressive detection rates ($> 93\%$–96%) and near-perfect AUC scores; however, these models are trained and tested exclusively on

static offline datasets (principally the SGCC repository or synthetic benchmark attack logs). They ignore real-world edge phenomena: analog noise, sensor non-linearities, temperature drift, sampling jitter, and network packet dropouts. On the other hand, empirical IoT hardware implementations [12], [13] utilize low-cost microcontrollers (ESP32) but rely entirely on crude static thresholds or manual visual inspection, lacking advanced statistical learning, feature engineering, or load forecasting.

B. High False Positive Rates and Lack of Safe Actuation Protocols

In real distribution grids, disconnecting power to a legitimate consumer due to a model false positive causes severe operational, legal, and reputational damage. Classifiers in literature often accept false positive rates between 3% and 7% [1], [3]. If a utility meter were configured to immediately cut power based on a single classification window output ($P(\text{theft}) > 0.5$), hundreds of honest households would be wrongly disconnected daily. Prior studies completely lack multi-stage persistence rules, safety timeouts, consumer alert stages, or graceful manual overrides before physical breaker/relay actuation.

C. Absence of Dual-Layered Detection Synergy

Supervised models [1], [3], [4] depend fundamentally on the availability of accurate, historical theft ground-truth labels. In new smart meter installations, historical labels do not exist. Conversely, unsupervised models [9], [10] operate without labels but suffer from higher false positive rates and cannot easily classify the specific mechanism of theft. Existing literature fails to combine unsupervised anomaly screening (which isolates novel deviations) with supervised gradient boosting into a unified pipeline where unsupervised scoring filters the input stream and bootstraps supervised verification.

D. Omission of Baseline Forecasting in Fraud Identification

Theft detection and demand forecasting are traditionally treated as completely disjoint problems. Forecasters [11] focus on utility load scheduling, while theft detectors [1], [2], [4] analyze consumption patterns in isolation. However, an accurate baseline forecast provides the exact dynamic expected-usage envelope against which anomalous drops should be judged. An unexpected 40% consumption drop during an extreme heatwave (when the forecasting model predicted peak cooling demand) is a far stronger indicator of theft than a drop during mild weather. Existing literature has not integrated Prophet-based baseline residual modeling directly into the theft decision engine.

E. How VoltIQ Bridges These Gaps

The proposed **VoltIQ** framework directly resolves these limitations by:

1) Integrating calibrated analog sensing (ZMPT101B and SCT-013 with ESP32) with offline microSD buffering to guarantee telemetry fidelity.
2) Enforcing a k-window persistence confirmation rule ($A(t) \geq \tau_a$ and $P(t) \geq \tau_p$ for k consecutive intervals) before actuating the cutoff relay, eliminating single-sample false trips.
3) Fusing unsupervised Isolation Forest screening, SMOTE-balanced XGBoost classification, and Prophet-BO-XGBoost baseline forecasting into an explainable, closed-loop dashboard architecture.

V. Proposed Work: The VoltIQ Framework

To address the identified research gaps, the VoltIQ framework is synthesized as a multi-tier, hardware-integrated cyber-physical architecture.

A. System Architecture and Data Pipeline

The VoltIQ system operates across five coordinated layers:

1) **Edge Sensing Layer:** An ESP32 microcontroller continuously samples an isolated ZMPT101B voltage transformer and a split-core SCT-013-030 current transformer at 1 kHz. Software RMS calculation produces V_{rms}, I_{rms}, and apparent power S. A DS3231 real-time clock ensures microsecond-accurate time stamps. Telemetry is saved locally to a circular microSD card buffer during network outages and simultaneously pushed over Wi-Fi/MQTT.
2) **Feature Engineering Engine:** Computes statistical features (rolling mean, standard deviation, peak-to-average ratio) across 15-minute and 24-hour windows, incorporating time-of-day and cyclical calendar flags [1], [6].
3) **Hybrid Forecasting Layer (Prophet-XGBoost):** Facebook Prophet extracts macro-level diurnal and weekly load periodicities, while XGBoost models residual non-linearities influenced by external weather variables [11]. The model outputs expected upper and lower consumption boundaries $[L(t), U(t)]$.
4) **Dual-Stage Detection Engine:**
   - *Unsupervised Screening:* An Isolation Forest computes anomaly score $A(t) \in [0, 1]$ based on isolation tree depth [9], [10].
   - *Supervised Classification:* If $A(t) \geq \tau_a$, the feature vector is processed by an XGBoost model trained with SMOTE resampling [1], [4], outputting theft probability $P(t) \in [0, 1]$.
5) **Safe Persistent Relay Actuation:** The relay cutoff is governed by:

$$\text{Theft}(t) = \begin{cases} 1, & \text{if } A(t) \geq \tau_a \text{ and } P(t) \geq \tau_p \quad \forall\, t \in [t-k+1, t] \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $k = 4$ consecutive 15-minute intervals (1-hour sustained fraud). Upon confirmation, the ESP32 energizes the optocoupled relay to disconnect the load, flags an audible alarm, dispatches an SMS alert, and logs an immutable tamper record to the Streamlit monitoring dashboard.

VI. Conclusion

This review paper surveyed thirteen peer-reviewed studies across IEEE, ScienceDirect, IET, and Springer, systematically evaluating developments in IoT smart metering, supervised theft detection, class imbalance handling, unsupervised anomaly screening, and hybrid consumption forecasting. Our comparative analysis demonstrated that existing algorithms remain disconnected from edge hardware, while low-cost IoT meters lack automated predictive analytics and safe actuation safeguards. The proposed **VoltIQ** framework unifies calibrated ESP32 sensing, Prophet-BO-XGBoost load baseline forecasting, Isolation Forest screening, SMOTE-boosted gradient boosting, and a k-window persistent relay actuation rule, providing a robust, explainable, and practical foundation for the implementation phase of our Final Year Project.

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TABLE I
COMPARATIVE ANALYSIS OF THE SURVEYED RESEARCH PAPERS

Paper Title	Authors	Year	Methodology / Technique	Key Findings	Advantages	Limitations
Energy Theft Detection Using Gradient Boosting Theft Detector [1]	R. Punmiya and S. Choe	2019	Gradient Boosting (XGBoost, LightGBM, CatBoost) with feature engineering.	XGBoost achieved highest AUC (> 0.92) and detection rate ($> 90\%$) with lower FPR than SVM.	Prunes irrelevant features; low computational complexity; models non-linearities.	Evaluated purely on offline data; lacks physical hardware or cutoff relay.
Wide & Deep CNN for Electricity-Theft Detection [2]	Z. Zheng et al.	2018	Wide & Deep CNN mapping 2D load matrices with cross-entropy loss.	Captured 2D temporal-periodic patterns; achieved AUC of 0.88 on SGCC dataset.	Eliminates manual feature engineering; captures subtle gradual tapering.	High compute demand; black-box nature lacks explainability; no edge device.
Using ML ensemble method for detection of energy theft [3]	A. I. Kawoosa et al.	2023	XGBoost, Random Forest, AdaBoost; 6 synthetic attack types.	XGBoost achieved 96% detection rate and 3% false positive rate across loads.	Robust against multimodal mathematical theft attacks; high precision.	Synthetic attacks may not match physical tampering behavior; regional specificity.
Combined Deep Learning and Ensemble Approach [4]	Z. Aslam et al.	2020	GRU + Ensemble Classifiers with SMOTE resampling.	SMOTE raised minority recall from 42% to 89%; overall accuracy reached $> 93\%$.	Resolves severe class imbalance; captures long-term sequential dependencies.	High training overhead; lacks closed-loop hardware or edge feasibility tests.
Robust Electricity Theft Detection Against Data Poisoning [5]	A. Takiddin et al.	2021	Autoencoder with Attention (AEA) + GRU against poisoning.	Generalized models exhibit higher resilience; reconstruction error detects spoofing.	Robust against label-flipping and telemetry poisoning; resilient under attack.	Complex deep network requiring server-grade GPU hardware; untested on edge.
Practical feature-engineering framework for theft detection [6]	R. Razavi et al.	2019	Statistical & behavioral feature engineering on AMI data.	Behavioral features (entropy, ratios) drastically boost fraud separability.	Drastically reduces raw time-series noise; yields interpretable metrics.	Algorithmic study; lacks automated relay actuation or real-time online testing.
Machine learning for NTL detection in electricity [7]	J. Pereira and F. Saraiva	2021	Random Forest, Extra Trees; customer profile segmentation.	Extra Trees achieved $> 95\%$ accuracy; customer-specific models beat global models.	Proves customer-segmentation necessity; evaluated on real billing data.	Residential consumers exhibit highest false alarms; lacks sensor calibration.
Electricity Theft Detection: CNN-LSTM Based Approach [8]	M. N. Hasan et al.	2019	Hybrid CNN-LSTM with SMOTE oversampling.	Attained F1-score of 0.89 and high AUC under severe class imbalance.	Simultaneously captures spatial local features and temporal sequential correlations.	Computationally intensive; recurrent layers suffer gradient degradation on long series.
Abnormal Electricity Detection: Canopy-Kmeans + iForest [9]	Y. Wang et al.	2024	Canopy-Kmeans clustering + Isolation Forest (iForest).	Successfully identified anomalous consumption without historical fraud labels.	Fully unsupervised; no reliance on scarce ground truth; high throughput.	Heuristic isolation threshold can miss fractional, stealthy bypass taps.
Anomaly Prediction in Consumption Using ML [10]	R. ELhadad et al.	2022	Isolation Forest labeling + Decision Tree & Random Forest.	Unsupervised auto-labeling enabled supervised forecasting of anomaly sequences.	Solves ground-truth scarcity; lightweight tree models suitable for fast inference.	Labels generated by iForest lack physical validation; sensitive to noise.
Short-Term Load Forecasting: Prophet-BO-XGBoost [11]	S. Zeng et al.	2025	Prophet + Bayesian Optimization + XGBoost.	Prophet handles trends while BO-XGBoost fits non-linear residuals accurately.	Additive seasonal decomposition paired with gradient-boosted residual regression.	Tested on aggregated grid loads; single-household erratic behavior untested.
IoT Based Smart Energy Meter with Tamper Proof [12]	S. S. Shinde et al.	2021	ESP32 microcontroller, Current Transformers, Relays, Cloud.	Dual-point differential current sensing accurately flagged illegal line bypasses.	Low-cost hardware; integrates physical relay load-shedding and cloud reporting.	Relies purely on static differential thresholds; lacks ML and forecasting.
Enhanced Low Cost Smart Energy Meter with Theft Detection [13]	T. Jayasree et al.	2023	ESP32, ZMPT101B voltage, ACS712 current, MQTT/HTTP.	Successfully streamed RMS voltage, current, and energy; flagged line breaks.	Low-cost IoT design; proves basic telemetry streaming feasibility.	High ADC error margins without calibration; lacks ML classification engine.